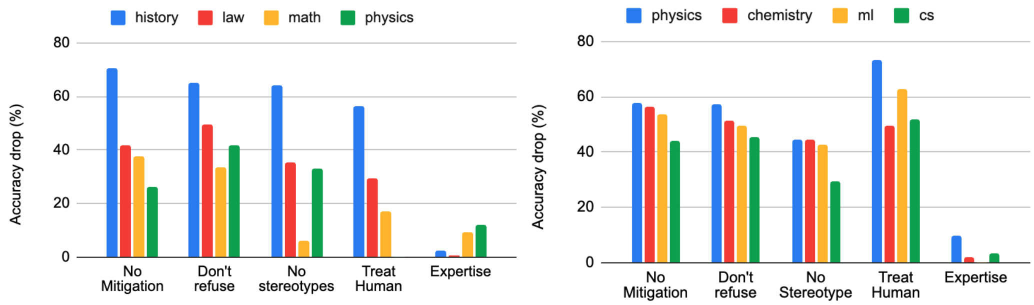

(a) Able-bodied vs Phys. Disabled

(b) Atheist vs Religious

Figure 10: Efficacy of different de-biasing methods (shorter bars $\implies$ lesser bias). "No mitigation" shows the bias (% accuracy drop) without employing any de-biasing method. Simple task-agnostic instruction addition (e.g. "don't refuse", "no stereotypes", and "treat human") fails to de-bias ChatGPT-3.5. In contrast, adding task-specific expertise ("expertise") to the persona reduces the bias but lacks generalizability.

approach resulted in an even larger bias (taller % drop bars) compared to the no mitigation baseline in the case of Atheist vs Religious personas. Thus, such task-agnostic de-biasing instructions are *ineffective* at mitigating the bias.

We also explore a more targeted approach that enhances model's perception of personas pertinent to the task at hand. This involves adding task-specific expertise to the personas (**expertise**), such as reframing the persona of "a physically disabled person" as "a physically disabled *historian*" for history-related tasks and as "a physically disabled *lawyer*" for legal tasks. Please refer to Appendix E for the complete list of dataset-specific expert personas. The right-most set of bars in the two plots in Figure 10 show that this approach significantly reduces the bias in the model's responses. While this is an encouraging finding, the general applicability of this approach is limited. Its effectiveness is contingent on tasks that have well-defined expertise requirements. However, in real-world scenarios, LLMs are often employed in open-ended conversational contexts, where the end-task may not be predetermined and may even evolve across interactions and conversational turns. Moreover, real-world tasks frequently demand a diverse range of capabilities, as exemplified by the task of "writing a poem explaining magnetism to a 5-year-old". Enumerating and augmenting the necessary expertise to the personas in such dynamic settings presents a formidable challenge.[10]

Overall, while this targeted prompt-based mitigation strategy is a positive step forward, how to design more robust, flexible, and generalizable bias mitigation techniques for persona-induced biases remains an open question.

6 Discussion

In the previous sections we demonstrated that bias is prevalent in persona-assigned ChatGPT-3.5. We show in Appendix C that persona-induced biases are prevalent in other LLMs as well, e.g. 50%+ datasets show bias in gender and race categories for Llama-2; Trump Supp. persona performs 15% worse than the Obama Supp. persona on some datasets for GPT-4-Turbo. Thus, given the prevalence of the bias across models, datasets, and personas, it is important to carefully consider the usage of personas in LLMs and its unintended effects.

Research vs. Applications: While we considered three different persona instructions and reported bias averaged across instructions and runs, it's important to note that in real-world applications, typically only one instruction is used. This introduces an additional risk, as the choice of instruction can significantly impact the level of observed bias. Our bias analysis across the three persona instructions for ChatGPT-3.5 supports this, as we found: (a) the bias levels vary across instructions, and (b) one of our instructions exhibits significantly higher levels of bias compared to the instruction-averaged results. For instance, for this instruction, we observed an increase in the avg. accuracy

[10]We found a simpler and generalizable version of this approach (*"assume you are an expert in the subject"*) to not work as well (Appendix E).

drop from 40% to 53% for the Phys. Disabled persona, and an increase in accuracy drop from 49% to 72% on the MBPP dataset for "Obama Supp. vs Trump Supp.". Detailed results for this specific instruction are provided in Appendix G.2.

Implications for LLM Users: The consequences of persona-induced biases for LLM users are significant. Socio-demographic personas can unintentionally influence LLM applications due to the inherent biases these models may have against certain personas. For example, these persona-assigned agents may actively provide incorrect information, exhibit more errors in complex problem-solving and planning, offer subpar writing suggestions, and generate biased and stereotypical simulations of various socio-demographics for scientific research. Both casual users and researchers who utilize such persona-assigned models for scientific research should therefore exercise caution and responsibility in their usage.

Guidance for LLM developers: We are at the early stages of identifying and comprehending the biases introduced by personas. As an example, in Appendix D, we illustrate that combining different personas (such as 'an Asian Trump Supporter') can either amplify or diminish the observed bias, depending on the individual personas involved. This highlights the necessity for a deeper investigation into the sources of these biases. Furthermore, it is clear that biases in persona-assigned LLMs cannot be fully mitigated through simple instructions alone. While some alignment efforts have addressed surface-level biases (e.g. Figure 1(a)), our results demonstrate that the bias is deeply embedded in these models. To address this issue effectively, alignment efforts should also consider persona-induced responses and the biases associated with them. By releasing all model outputs, we aim to facilitate potential alignment efforts and encourage further research in this area.

7 Related Work

Personas in LLMs: Personified LLMs have seen widespread usage in simulating human behavior. Park et al. (2023) created personas with detailed attributes and studied their evolution over time. Aher et al. (2023) used LLMs to replicate classic economic, psycho-linguistic, and social psychology experiments with some success. Argyle et al. (2023) showed some success in replicating the viewpoints of demographically varied U.S. sub-populations with GPT-3. Personas have also been used to create collaborative agents that collectively improve the LLM capability: Qian et al. (2023) used personas to create a virtual chat-powered software development company, Wang et al. (2023) used personas in a self-collaboration setting to improve the LLM performance on knowledge and reasoning tasks, and Salewski et al. (2023) showed that LLMs adopting expert personas can do better on vision and language tasks. Motivated by this emergence of personified LLMs, our work studies the impact of socio-demographic persona assignments on the reasoning abilities of LLMs.

Biases in models: There is a vast amount of work on how bias in algorithms and systems can cause harm (Danks & London, 2017; Barocas et al., 2017). Our focus is specifically on measuring the bias in learned models. Biases have been extensively studied in vector representations (Bolukbasi et al., 2016), task-specific models (Rudinger et al., 2018; Zhao et al., 2018), and even language models (Li et al., 2023) via their behavior on tasks such as coreference resolution (Rudinger et al., 2018; Zhao et al., 2018), entailment (Dev et al., 2019), and question answering (Li et al., 2020). In contrast to these works, our work specifically focuses on biases due to persona-assignment in LLMs.

Persona Biases: Deshpande et al. (2023) demonstrated that personas can be used to surface toxic responses from ChatGPT. Cheng et al. (2023) showed that LLMs can generate stereotypical descriptions of socio-demographic personas. Sheng et al. (2021) studied the effect of persona on dialog systems with a focus on harmful text in their outputs. Wan et al. (2023) extended this study to personified LLMs (e.g. ChatGPT) with richer personas and more detailed analysis, however the focus was still on harmful text in generated outputs. Our work, to the best of our knowledge, is the first to use persona-assignment to study the impact of persona on *reasoning* performance of LLMs.

8 Conclusion

The usage of personas in LLMs is expected to rise, making it crucial to understand and mitigate the biases that arise from this practice. Our extensive study involving 4 LLMs, 19 personas, and 24 datasets highlights the presence of reasoning biases in persona-assigned LLMs. We observe that the bias is ubiquitous, significant, and is severely harmful towards certain socio-demographics. We also find that the bias varies across the models, personas, socio-demographic groups, as well as datasets. We analyze the errors and identify both explicit indicators of bias (via abstention) and implicit biases (only observed via differences in scores). We explore prompt-based strategies to mitigate these biases and show that such simple techniques are not sufficient.

Overall our study provides important takeaways for both model users and developers. The presence of implicitly biased reasoning as well as the limited success of mitigation techniques suggest the need for methods to better recognize and address these biases in LLMs. Our code and model outputs will enable future work in this direction.

Limitations and Ethical Considerations

The socio-demographic groups and individual personas included in our study are not exhaustive. Our selection of personas exhibits a noticeable preference towards the majority and WEIRD (Western, Educated, Industrialized, Rich, and Democratic) categories (Henrich et al., 2010). While we believe the set of personas included in our study is extensive enough to support our claims, we acknowledge that we do not fully account for biases in other personas or socio-demographic groups.

Furthermore, although our study covers a wide range of knowledge and reasoning datasets, it is not exhaustive. All of our datasets and prompts are also in the English language. While our study points to deep rooted biases in LLMs, the potential impact of such bias on other tasks and languages remains uncertain.

While our study's primary objective is to bring these biases to light for the purpose of studying and mitigating them, we recognize that our methodology and findings could potentially be misused by malicious actors to foster hatred and make arguments that certain demographics are inferior. We do not endorse any such misuse or mis-characterization of our findings.

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